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POWER SUPPLY SYSTEM AND ELECTRONIC DEVICE

Abstract

The application discloses a power supply system and an electronic device. The power supply system includes a rectifying circuit module, a DC connecting bus module and a voltage output circuit module. The rectifying circuit module is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module is connected with the rectifying circuit module to store the DC current and output a DC voltage; the voltage output circuit module is connected with the DC connecting bus module to convert the DC voltage of the DC connecting bus module to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims the benefit of priority to Chinese Patent Application No. 202410257621.1, filed on Mar. 6, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This application pertains to a power supply technical field, in particular to a power supply system and an electronic device.

TECHNICAL BACKGROUND

[0003] With the development of power electronic devices and their control technologies, the requirements of electronic devices for the quality of input power and the stability of voltage of the power supply system become higher.

[0004] As for electronic devices, taking medical devices as an example, medical imaging devices are applied to obtaining images of human tissues in the medical field. It includes many subsystems that require different types and levels of voltage, including AC voltage, DC voltage, high voltage, low voltage, constant voltage, gradient voltage, etc. In addition, the power supply systems of medical imaging devices are also required to process many voltage levels and frequencies in different geographical regions.

[0005] In the related art, to meet the above-mentioned requirements, the power supply systems of traditional medical imaging devices shall be regulated to match different input voltage levels and frequencies (such as voltage standards in different countries and 50 Hz or 60 Hz). Such transformers often need many input taps and the iron core materials shall match with different voltages and frequencies, which inevitably causes inconvenience and unnecessary wastes to system design. In addition, due to the lack of effective feedback regulation control mechanisms, the output voltage (AC or DC) of this power supply system based on traditional power transformers fluctuates with changes in the input voltage, causing adverse effects on medical imaging devices.

SUMMARY

[0006] This application proposes a power supply system and an electronic device with a view to solving the above problems.

[0007] To solve the above technical problems, this application proposes a technical scheme that is a power supply system which includes a rectifying circuit module, a DC connecting bus module and a voltage output circuit module. The rectifying circuit module is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module is connected with the rectifying circuit module to store the DC current and output a DC voltage; and the voltage output circuit module is connected with the DC connecting bus module to convert the DC voltage of the DC connecting bus module to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it.

[0008] In some embodiments, the DC connecting bus module includes a DC connecting bus and an input interface of the external power supply; the DC connecting bus is connected with the rectifying circuit module and the voltage output circuit module to store the DC current and output a DC voltage; the input interface of the external power supply is connected with the DC connecting bus to connect with other external DC power supplies.

[0009] In some embodiments, the rectifying circuit module includes a three-phase power factor

correction circuit and a control circuit; the three-phase power factor correction circuit corrects and rectifies the AC current of the external power supply to form a DC current; the control circuit is connected with the three-phase power factor correction circuit to realize the control of the three-phase power factor correction circuit by a digital signal processing method.

[0010] In some embodiments, the voltage output circuit module includes a three-phase AC voltage output module, a single-phase AC voltage output module and a DC voltage output module; the three-phase AC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a three-phase AC voltage and output it; the single-phase AC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a single-phase AC voltage and output it; and the DC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a target DC voltage and output it.

[0011] The DC connecting bus module includes the first DC connecting bus, the second DC connecting bus and an input interface of the external power supply, and the rectifying circuit module includes the first three-phase power factor correction circuit, the second three-phase power factor correction circuit and a control circuit; the first three-phase power factor correction circuit is connected with the first DC connecting bus, and the three-phase AC voltage output module and the single-phase AC voltage output module are connected with the first DC connecting bus; and the second three-phase power factor correction circuit is connected with the second DC connecting bus and the DC voltage output module is connected with the second DC connecting bus.

[0012] In some embodiments, the three-phase AC voltage output module includes a three-phase DC to AC converter, the first AC filter and a three-phase AC output circuit; the three-phase DC to AC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a three-phase AC voltage; the first AC filter is connected with the three-phase DC to AC converter to filter the three-phase AC voltage; and the three-phase AC output circuit is connected with the first AC filter to output the three-phase AC voltage.

[0013] In some embodiments, the single-phase AC voltage output module includes a single-phase DC to AC converter, the second AC filter and a single-phase AC output circuit; the single-phase DC to AC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a single-phase AC voltage; the second AC filter is connected with the single-phase DC to AC converter to filter the single-phase AC voltage; and the single-phase AC output circuit is connected with the second AC filter to output a single-phase AC voltage.

[0014] In some embodiments, the DC voltage output module includes a DC to DC converter, a DC filter and a DC output circuit; the DC to DC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a target DC voltage; the DC filter is connected with the DC to DC converter to filter the target DC voltage; and the DC output circuit is connected with the DC filter to output the target DC voltage.

[0015] In some embodiments, the power supply system also includes an AC input and protection circuit module which includes an AC input sub-module, an isolation switch sub-module, a surge protection sub-module, an electromagnetic interference sub-module and an input current limiting sub-module; the AC input sub-module is connected with the external power supply to receive the AC input of the external power supply; the isolation switch sub-module is connected with the AC input sub-module to protect the power supply system to operate normally; the surge protection sub-module is connected with the isolation switch sub-module to prevent the damage caused by the surge voltage or current of the external power supply; the electromagnetic interference sub-module is connected with the surge protection sub-module to suppress electromagnetic interference; and the input current limiting sub-module is connected with the electromagnetic interference sub-module to limit the AC current of the external power supply.

[0016] In some embodiments, the power supply system also includes a DC/DC HF isolation transformer module which is connected with the DC connecting bus module and the voltage output circuit module to regulate the DC voltage of the DC connecting bus module to the DC voltage level

required by the voltage output circuit module.

[0017] In some embodiments, the power supply system also includes an output monitoring module and a grounding failure protection module; and the output monitoring module is connected with the DC/DC HF isolation transformer module to monitor the voltage and current outputted by the DC/DC HF isolation transformer module and the grounding failure protection module is connected with the voltage output circuit module to monitor the grounding failure of the DC voltage outputted by the voltage output circuit module.

[0018] In some embodiments, the power supply system also includes a central protection and failure monitoring processing module which includes a central management sub-module and an external communication interface. The central management sub-module is used to monitor the status and safety of the power supply system in real time and quickly and accurately execute protection tasks in a failure; and the external communication interface is connected with the central management sub-module to realize the communication between the central management sub-module and the external device.

[0019] In some embodiments, the power supply system also includes an internal auxiliary power module which is connected with the central management sub-module to provide a safe internal power guarantee to the central management sub-module.

[0020] To solve the above technical problems, this application discloses a technical scheme that is an electronic device which includes any power supply system mentioned above.

[0021] The beneficial effects of this application are as follows: different from the related art, the power supply system in this application includes a rectifying circuit module, a DC connecting bus module and a voltage output circuit module; the rectifying circuit module is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module is connected with the rectifying circuit module to store the DC current and output a DC voltage; and the voltage output circuit module is connected with the DC connecting bus module to convert the DC voltage of the DC connecting bus module to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it. The power supply system in this application takes the DC connecting bus module as a transfer module to ensure stability and regulation of AC voltage and/or DC voltage outputted by the voltage output circuit module without the influence of fluctuation on the AC side of the rectifying circuit module. In addition, in the application, the voltage output circuit module converts the DC voltage of the DC connecting bus module to AC voltage and/or DC voltage, can replace the multiple-tap structure of traditional power supply systems and can save costs.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] The following figures are included in the specification and constitute a part of the specification. These figures show the embodiments in line with this application and are attached to the specification to describe the technical scheme of this application.

[0023] FIG. 1 illustrates a structure schematic diagram of the first embodiment of the power supply system in this application;

[0024] FIG. 2 illustrates a structure schematic diagram of the second embodiment of the power supply system in this application;

[0025] FIG. 3 illustrates a circuit structure schematic diagram of the first embodiment of the three-phase power factor correction circuit in this application;

[0026] FIG. 4 illustrates a circuit structure schematic diagram of the second embodiment of the three-phase power factor correction circuit in this application;

[0027] FIG. 5 illustrates a circuit structure schematic diagram of the third embodiment of the three-

phase power factor correction circuit in this application;
[0028] FIG. **6** illustrates a circuit structure schematic diagram of the first embodiment of the three-phase DC to AC converter in this application;
[0029] FIG. **7** illustrates a circuit structure schematic diagram of the second embodiment of the three-phase DC to AC converter in this application;
[0030] FIG. **8** illustrates a circuit structure schematic diagram of the third embodiment of the three-phase DC to AC converter in this application;
[0031] FIG. **9** illustrates a circuit structure schematic diagram of the first embodiment of the single-phase DC to AC converter in this application;
[0032] FIG. **10** illustrates a circuit structure schematic diagram of the second embodiment of the single-phase DC to AC converter in this application;
[0033] FIG. **11** illustrates a circuit structure schematic diagram of the third embodiment of the single-phase DC to AC converter in this application;
[0034] FIG. **12** illustrates a structure schematic diagram of the third embodiment of the power supply system in this embodiment;
[0035] FIG. **13** illustrates a circuit structure schematic diagram of the first embodiment of the DC/DC HF isolation transformer in this application;
[0036] FIG. **14** illustrates a circuit structure schematic diagram of the second embodiment of the DC/DC HF isolation transformer in this application;
[0037] FIG. **15** illustrates a circuit structure schematic diagram of the third embodiment of the DC/DC HF isolation transformer in this application;
[0038] FIG. **16** illustrates a structure schematic diagram of the fourth embodiment of the DC/DC HF isolation transformer in this application; and
[0039] FIG. **17** illustrates a structure schematic diagram of an embodiment of the electronic device in this application.

DESCRIPTION OF REFERENCE NUMERALS

[0040] **100**—Power supply system;
[0041] **10**—Rectifying circuit module;
[0042] **11**—Three-phase power factor correction circuit;
[0043] **12**—Control circuit;
[0044] **13**—The first three-phase power factor correction circuit;
[0045] **14**—The second three-phase power factor correction circuit;
[0046] **20**—DC connecting bus module;
[0047] **21**—DC connecting bus;
[0048] **22**—Input interface of the external power supply;
[0049] **23**—The first DC connecting bus;
[0050] **24**—The second DC connecting bus;
[0051] **30**—Voltage output circuit module,
[0052] **31**—Three-phase AC voltage output module;
[0053] **311**—Three-phase DC to AC converter;
[0054] **312**—The first AC filter;
[0055] **313**—Three-phase AC output circuit;
[0056] **32**—Single-phase AC voltage output module;
[0057] **321**—Single-phase DC to AC converter;
[0058] **322**—The second AC filter;
[0059] **323**—Single-phase AC output circuit;
[0060] **33**—DC voltage output module;
[0061] **331**—DC to DC converter;
[0062] **332**—DC filter;
[0063] **333**—DC output circuit;

[0064] **40**—AC input and protection circuit module;
[0065] **41**—AC input sub-module;
[0066] **42**—Isolation switch sub-module;
[0067] **43**—Surge protection sub-module;
[0068] **44**—Electromagnetic interference sub-module;
[0069] **45**—Input current limiting sub-module;
[0070] **50**—DC/DC HF isolation transformer module;
[0071] **60**—Output monitoring module;
[0072] **70**—Grounding failure protection module;
[0073] **80**—Central protection and failure monitoring processing module;
[0074] **81**—Central management sub-module;
[0075] **82**—External communication interface;
[0076] **90**—Internal auxiliary power module;
[0077] **200**—Electronic device.

DETAILED DESCRIPTION

[0078] The embodiments of the technical scheme in this application will be detailed below with reference to drawings. The following embodiments are only presented to provide a clearer explanation of the technical scheme of this application, and are therefore only used as examples and cannot limit the scope of protection of this application.

[0079] Unless otherwise defined, all technical and scientific terms mentioned in this application contain the same meanings as those commonly understood by person skilled in the art of this application; the terms mentioned in this application are only provided for describing the specific embodiments and are not intended to limit this application; the terms “comprising” and “possessing” in the specification, claims and drawings in this application and any variations thereof are intended to cover non-exclusive inclusion.

[0080] In the description of embodiments in this application, the technical terms “the first,” “the second,” etc. are only intended to distinguish different objects, but are not regarded as the indication or implication of relative importance or the implied indication of the quantity, specific order or primary relationship of the indicated technical features. Among the description of embodiments in this application, the term “many” means more than two unless otherwise specified.

[0081] The “embodiment” in this application means that specific features, structures or characteristics described in conjunction with the embodiment can be included in at least one embodiment in this application. The embodiments in different positions in the specification are not necessarily referring to the same embodiment and are not independent or alternative embodiments from other embodiments. The persons skilled in the art explicitly and implicitly understand that embodiments mentioned in this application can be combined with other embodiments.

[0082] With the development of power electronic devices and their control technologies, the requirements of electronic devices for the quality of input power and the stability of voltage of the power supply system become higher.

[0083] As for electronic devices, taking medical devices as an example, medical imaging devices are applied to obtaining images of human tissues in the medical field. It includes many subsystems that require different types and levels of voltage, including AC voltage, DC voltage, high voltage, low voltage, constant voltage, gradient voltage, etc. In addition, the power supply systems of medical imaging devices are also required to process many voltage levels and frequencies in different geographical regions and to provide the current isolation and protection equipment for loads and AC power supply to avoid being influenced by grounding failure.

[0084] In the related art, to meet the above mentioned requirements, the power supply systems of traditional medical imaging devices shall be regulated to match different input voltage levels and frequencies (such as voltage standards in different countries and 50 Hz or 60 Hz). Such transformers often need many input taps and the iron core materials match with different voltages

and frequencies, which inevitably causes inconvenience and unnecessary wastes to system design. In addition, due to the lack of effective feedback regulation control mechanisms, the output voltage (AC or DC) of this power supply system based on traditional power transformers fluctuates with changes in the input voltage, causing adverse effects on medical imaging devices.

[0085] Besides, the current power supply systems of medical devices also have other disadvantages, such as large volume, weight and no-load losses; the primary and secondary sides of the transformer of the power supply system are prone to be influenced by failures and interference, and unwanted characteristics on the input side (such as voltage drop) will be reflected in the output waveform; and in addition, the frequency of the AC output voltage is constant and can not be regulated. Therefore, the power supply system has no variable frequency speed regulation function. [0086] To solve the above problems, this application firstly proposes a power supply system. Please refer to FIG. 1. FIG. 1 illustrates a structure schematic diagram of the first embodiment of the power supply system in this application. As shown in FIG. 1, in this embodiment, the power supply system **100** includes a rectifying circuit module **10**, a DC connecting bus module **20** and a voltage output circuit module **30**.

[0087] As shown in FIG. 1, the rectifying circuit module **10** is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module **20** is connected with the rectifying circuit module **10** to store the DC current and output a DC voltage; and the voltage output circuit module **30** is connected with the DC connecting bus module **20** to convert the DC voltage of the DC connecting bus module **20** to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it.

[0088] In this embodiment, the rectifying circuit module **10** can include a three-phase power factor correction circuit which includes a three-phase rectifying circuit module and an active power factor correction circuit. This circuit plays three roles in this embodiment. The first role is to rectify the three-phase AC of the external power supply to DC, form a DC bus and regulate and control the bus voltage. The second role is to correct the input current on the AC side of the external power supply to be close to sine so as to improve the power factor. The third role is that in the case of asymmetric three-phase input from the external power supply, a phase locked loop in the rectifying circuit module **10** can decompose the positive and negative orders of the three-phase input voltage to ensure that the current contains no negative order components and to improve power quality.

[0089] The DC connecting bus module **20** includes a DC connecting bus which can be regarded as a connection link of input and output of the whole power supply system **100** of the embodiment, as a transfer station, a buffer zone or a reservoir for energy conversion and as a public interface of the power supply system **100**.

[0090] In this embodiment, the voltage output circuit module **30** includes at least any of a three-phase AC voltage output module, a single-phase AC voltage output module or a DC voltage output module. The three-phase AC voltage output module mainly modulates and converts the DC voltage to a three-phase AC voltage; the single-phase AC voltage output module mainly modulates and converts the DC voltage to a single-phase AC voltage; the DC voltage output module mainly converts the DC voltage to a target DC voltage further so as to accurately output the target DC voltage.

[0091] Different from the related art, the power supply system **100** in this application includes a rectifying circuit module **10**, a DC connecting bus module **20** and a voltage output circuit module **30**. The rectifying circuit module **10** is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module **20** is connected with the rectifying circuit module **10** to store the DC current and output a DC voltage; and the voltage output circuit module **30** is connected with the DC connecting bus module **20** to convert the DC voltage of the DC connecting bus module **20** to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it. The

power supply system **100** in this application takes the DC connecting bus module **20** as the transfer module to ensure stability and regulation of AC voltage and/or DC voltage outputted by the voltage output circuit module **30** without the influence of fluctuation on the AC side of the rectifying circuit module **10**. In addition, in the application, the voltage output circuit module **30** converts the DC voltage of the DC connecting bus module **20** to AC voltage and/or DC voltage, can replace the multiple-tap structure of traditional power supply systems and can save costs.

[0092] In some embodiments, refer to FIG. 2. FIG. 2 illustrates a structure schematic diagram of the second embodiment of the power supply system in this application. As shown in FIG. 2, the DC connecting bus module **20** in the embodiment includes a DC connecting bus **21** and an input interface of the external power supply **22**. The DC connecting bus **21** is connected with the rectifying circuit module **10** and the voltage output circuit module **30** to store the DC current and output a DC voltage; and the input interface of the external power supply **22** is connected with the DC connecting bus **21** to connect with other external DC power supply.

[0093] In some embodiments, in this embodiment, the DC connecting bus module **20** is set with the input interface of the external power supply **22** to easily access any other external DC power supply or load directly or indirectly so as to provide more power selection and redundant design. In addition, other external DC power supply can be accessed by the input interface of the external power supply **22**. The DC connecting bus **21** can easily convert to the required power output type. In theory, the input interface of the external power supply **22** can couple any DC power supply, DC load and energy storage device, such as photovoltaic solar energy, energy storage capacitors and battery modules. In this way, power supply forms other than AC power grid can be provided to achieve the mobility of the power supply system **100**.

[0094] In this embodiment, the DC connecting bus **21** is set as the connection link of input and output of the power supply system **100**, so the AC voltage and/or DC voltage outputted by the voltage output circuit module **30** can be stabilized and regulated in a convenient way without the influence of fluctuation on the AC side, such as asymmetric changes in input voltage, voltage drop or frequency instability. However, in this embodiment, all outputs are regulated and converted from DC voltage by the DC bus. The voltage of the DC bus is stabilized at a constant value by control. No matter how the peak and frequency of the input voltage change, the AC voltage and/or DC voltage outputted by the voltage output circuit module **30** can be stabilized and regulated by feedback control conversion with higher precision and more stable output. The output of traditional power supply systems varies with input fluctuations. The power supply system **100** in this application has advantages over traditional power supply systems because of its relative isolation and independent control of input and output.

[0095] In some embodiments, as shown in FIG. 2, in this embodiment, the rectifying circuit module **10** includes a three-phase power factor correction circuit **11** and a control circuit **12**. The three-phase power factor correction circuit **11** is provided to correct and rectify the AC current of the external power supply to form a DC current; and the control circuit **12** is connected with the three-phase power factor correction circuit **11** to realize the control of the three-phase power factor correction circuit **11** by a digital signal processing method.

[0096] As mentioned above, the three-phase power factor correction circuit **11** can not only correct and rectify the AC current of the external power supply to form a DC current, but also correct the input current on the AC side of the external power supply to be close to sine so as to improve the power factor. In addition, in the case of asymmetric three-phase input from the external power supply, a phase locked loop in the three-phase power factor correction circuit **11** can decompose the positive and negative orders of the three-phase input voltage to ensure that the current contains no negative order components and to improve power quality.

[0097] In some embodiments, refer to FIG. 3 to FIG. 5. FIG. 3 illustrates a circuit structure schematic diagram of the first embodiment of the three-phase power factor correction circuit in this application. FIG. 4 illustrates a circuit structure schematic diagram of the second embodiment of

the three-phase power factor correction circuit in this application. FIG. 5 illustrates a circuit structure schematic diagram of the third embodiment of the three-phase power factor correction circuit in this application. In this embodiment, the three-phase power factor correction circuit **11** can be set as a circuit topological structure of any structure in FIG. 3 to FIG. 5. In other embodiments, the three-phase power factor correction circuit **11** can be set as other circuit topological structures on the premise of having above functions, with no more restrictions here.

[0098] In this embodiment, although the three-phase voltage input from an external power supply shows a sinusoidal waveform, the input current waveform is non-sinusoidal due to rectification or non-linear loads. The power factor is very low. The power supply system **100** in this embodiment is set with the three-phase power factor correction circuit **11** to correct the input current waveform to be close to sine and in phase with voltage, which greatly improves the power factor to close to 1. That is to say, the three-phase power factor correction circuit **11** in this embodiment has both the function of converting AC to DC and stepping up and the function of correcting power factor. Although the three-phase voltage is sinusoidal, the three-phase input current may not be sinusoidal, which is caused by the different load types. Anyway, the control algorithms, such as phase locked loop and regulation methods, make the input current infinitely close to sine and in phase with voltage waveform. In this way, the power factor is close to 1 so as to achieve the purpose of correcting the power factor. Especially, in the case of asymmetric three-phase input voltage, the positive and negative order decomposition algorithm of the phase locked loop can ensure that the input current only has the positive order in-phase sine waveform and has no negative order component to further guarantee the correction of power factor.

[0099] In some embodiments, as shown in FIG. 2, in this embodiment, the voltage output circuit module **30** includes a three-phase AC voltage output module **31**, a single-phase AC voltage output module **32** and a DC voltage output module **33**. The three-phase AC voltage output module **31** is connected with the DC connecting bus module **20** to convert the DC voltage to a three-phase AC voltage and output it; the single-phase AC voltage output module **32** is connected with the DC connecting bus module **20** to convert the DC voltage to a single-phase AC voltage and output it; and the DC voltage output module **33** is connected with the DC connecting bus module **20** to convert the DC voltage to a target DC voltage and output it.

[0100] As mentioned above, the three-phase AC voltage output module **31** mainly regulates and converts the DC voltage to a three-phase AC voltage and provides a true neutral wire to meet the requirements of an effective grounding system. This part also has a voltage regulation and frequency feedback control function to accurately output the desired AC voltage; the single-phase AC voltage output module **32** mainly regulates and converts the DC voltage to a single-phase AC voltage. This part also includes voltage regulation and frequency feedback control function to accurately output the desired AC voltage; and in this embodiment, the DC voltage output module **33** can be set with the second level DC to DC conversion. According to a specific situation, the DC voltage output module **33** can be isolated or non-isolated. Its main function is to further convert the DC voltage to a target DC voltage. This part also has the voltage regulation feedback control function to accurately output a target DC voltage.

[0101] In other embodiments, the voltage output circuit module **30** can include at least any of a three-phase AC voltage output module **31**, a single-phase AC voltage output module **32** and a DC voltage output module **33**. That is to say, the voltage output circuit module **30** can be set based on the actual voltage. For example, if only AC voltage is needed, the voltage output circuit module **30** can only include a three-phase AC voltage output module **31** and a single-phase AC voltage output module **32**; and if only DC voltage is needed, the voltage output circuit module **30** can only include a DC voltage output module **33**.

[0102] In some embodiments, as shown in FIG. 2, in this embodiment, the three-phase AC voltage output module **31** includes a three-phase DC to AC converter **311**, the first AC filter **312** and a three-phase AC output circuit **313**. The three-phase DC to AC converter **311** is connected with the

DC connecting bus **21** to convert the DC voltage of the DC connecting bus **21** to a three-phase AC voltage; the first AC filter **312** is connected with the three-phase DC to AC converter to filter the three-phase AC voltage; and the three-phase AC output circuit **313** is connected with the first AC filter **312** to output the three-phase AC voltage.

[0103] In some embodiments, in this embodiment, refer to FIG. 6 to FIG. 8. FIG. 6 illustrates a circuit structure schematic diagram of the first embodiment of the three-phase DC to AC converter in this application. FIG. 7 illustrates a circuit structure schematic diagram of the second embodiment of the three-phase DC to AC converter in this application. FIG. 8 illustrates a circuit structure schematic diagram of the third embodiment of the three-phase DC to AC converter in this application. In this embodiment, the three-phase DC to AC converter can be set as the circuit topological structure of any structure in FIG. 6 to FIG. 8. In some embodiments, if the three-phase DC to AC converter **311** is set as the three-phase four-wire DC to AC converter in FIG. 8, it can provide a true neutral wire to realize the effective grounding protection function. In other embodiments, the three-phase DC to AC converter **311** can be set as other circuit topological structures on the premise of having above functions, with no more restrictions here.

[0104] In this embodiment, the three-phase DC to AC converter **311** is preferably set as a three-phase four-wire DC to AC converter which can effectively ground the system and copy with nonlinear and three-phase unbalanced loads. In the case of a single-phase grounding failure, the voltage of the other two phases will not increase to affect safety, the failure can be accurately judged and the phase can be quickly cut. In general, the traditional three-phase DC to AC converter **311** has three wires and has no neutral wire. In the case of a single-phase grounding failure, the phase voltage of a certain phase becomes zero and the phase voltage of other two phases also changes so that the voltage between phases is unchanged. Due to the lack of a neutral wire, the measured voltage is the phase voltage of three phases. In such a case, it is impossible to judge whether there is a single-phase grounding failure from the perspective of the voltage. The power supply system **100** can continue to operate, but an increase in phase voltage of other two phases can cause injury to operators and damage to device insulation. Therefore, the preferred configuration of the three-phase DC to AC converter **311** as a three-phase four-wire DC to AC converter to avoid any injury to operators and damage to device insulation. In addition, whether measuring the wire voltage or phase voltage of three phases, combined with current measurement, it is easy to judge the specific phase in failure, thus meeting the effective grounding requirements of the power supply system **100**.

[0105] In addition, if the three-phase DC to AC converter **311** is preferably set as a three-phase four-wire DC to AC converter, in this embodiment, the same effective grounding can be achieved by installing the fourth controllable switch arm to independently control the balance of midpoint voltage on the DC side through regulation. In the case of single-phase grounding failure, it plays the same effective grounding role as traditional power supply system **100**. Besides, if the three-phase loads are imbalanced, the neutral wire can bear this imbalanced zero sequence current to guarantee normal work of the three-phase inverter.

[0106] In this embodiment, the three-phase AC output circuit **313** is also set with proper measurement and feedback control device to easily realize the AC variable frequency speed regulation function, directly drive the three-phase induction motor, and realize soft start, speed control, constant torque control, constant power control, etc.

[0107] In some embodiments, as shown in FIG. 2, in this embodiment, the single-phase AC voltage output module **32** includes a single-phase DC to AC converter **321**, the second AC filter **322** and a single-phase AC output circuit **323**. The single-phase DC to AC converter **321** is connected with the DC connecting bus **21** to convert the DC voltage of the DC connecting bus **21** to a single-phase AC voltage; the second AC filter **322** is connected with the single-phase DC to AC converter to filter the single-phase AC voltage; and the single-phase AC output circuit **323** is connected with the second AC filter **322** to output a single-phase AC voltage.

[0108] In some embodiments, in this embodiment, refer to FIG. 9 FIG. 11. FIG. 9 illustrates a circuit structure schematic diagram of the first embodiment of the single-phase DC to AC converter in this application. FIG. 10 illustrates a circuit structure schematic diagram of the second embodiment of the single-phase DC to AC converter in this application. FIG. 11 illustrates a circuit structure schematic diagram of the third embodiment of the single-phase DC to AC converter in this application. In this embodiment, the single-phase DC to AC converter **321** can be set as a circuit topological structure of any structure in FIGS. 9-11. In other embodiments, the single-phase DC to AC converter **321** can be set as other circuit topological structures, such as the single-phase H5 inverter or single-phase H6 inverter. In addition, cascade or parallel connection can be used to increase the voltage grade and power output. To sum up, the single-phase DC to AC converter **321** only needs to have the above functions, with no more restrictions here.

[0109] In this embodiment, the single-phase AC output circuit **323** is also set with proper measurement and feedback control device to quickly and accurately output the desired single-phase AC voltage.

[0110] In some embodiments, as shown in FIG. 2, in this embodiment, the DC voltage output module **33** includes a DC to DC converter **331**, a DC filter **332**, and a DC output circuit **333**. The DC to DC converter **331** is connected with the DC connecting bus **21** to convert the DC voltage of the DC connecting bus **21** to a target DC voltage; the DC filter **332** is connected with the DC to DC converter **331** to filter the target DC voltage; and the DC output circuit **333** is connected with the DC filter **332** to output the target DC voltage.

[0111] In this embodiment, as mentioned above, the DC to DC converter **331** can be known as the second-level DC voltage control. Wherein, in this embodiment, the DC output circuit **333** is also set with proper measurement and feedback control devices to obtain a target DC voltage in a faster and more accurate way.

[0112] In some embodiments, refer to FIG. 12. FIG. 12 illustrates a structure schematic diagram of the third embodiment of the power supply system in this embodiment. In this embodiment, different from the embodiment in FIG. 2, the three-phase AC voltage output module **31** and the single-phase AC voltage output module **32** in this embodiment share the first DC connecting bus **23** and the first three-phase power factor correction circuit **13**; and the DC voltage output module **33** is set with the independent second DC connecting bus **24** and the second three-phase power factor correction circuit **14**.

[0113] As shown in FIG. 12, the DC connecting bus module **20** includes the first DC connecting bus **23**, the second DC connecting bus **24** and an input interface of the external power supply **22**. The rectifying circuit module **10** includes the first three-phase power factor correction circuit **13**, the second three-phase power factor correction circuit **14** and a control circuit **12**. The first three-phase power factor correction circuit **13** is connected with the first DC connecting bus **23** and the three-phase AC voltage output module **31** and the single-phase AC voltage output module **32** are connected with the first DC connecting bus **23**; the second three-phase power factor correction circuit **14** is connected with the DC connecting bus **24**; and the DC voltage output module **33** is connected with the second DC connecting bus **24**.

[0114] Wherein, both the first three-phase power factor correction circuit **13** and the second three-phase power factor correction circuit **14** are connected with the control circuit **12** and are controlled by the control circuit **12**. In this embodiment, the control circuit **12** can be set as the DSP control circuit **12**. The DSP control circuit **12** is a circuit that utilizes digital signal processing methods to control electronic devices or systems and a microprocessor for fast mathematical operations and algorithm execution. It is widely used in various electronic systems, including but not limited to switch power control, internal effector of audio equipment such as mixing consoles, communication equipment and industrial control systems.

[0115] In addition, as shown in FIG. 2, the DC/DC HF isolation transformer module **50** is set between the DC connecting bus module **20** and the DC voltage output module **33** of the power

supply system **100**. The DC/DC HF isolation transformer module **50** can regulate the DC voltage of the DC connecting bus module **20** to the DC voltage level required by the voltage output circuit module **30**. In this embodiment, as shown in FIG. **12**, because the DC voltage output module **33** is set with the independent second DC connecting bus **24** and the second three-phase power factor correction circuit **14**, the DC to DC converter **331** in the DC voltage output module **33** can be omitted. The DC/DC HF isolation transformer module **50** can directly convert the DC voltage of the DC connecting bus module **20** to a DC voltage level required by the DC voltage output module **33** without the need of setting the DC to DC converter **331** to realize the second-level DC voltage conversion.

[0116] Compared with the power supply system **100** in the embodiment in FIG. **2**, in this embodiment, the AC voltage output module and the DC voltage output module **33** are relatively independent and can be installed and replaced in an easier way.

[0117] In some embodiments, refer to FIG. **2** and FIG. **12**. Based on above embodiments, in this embodiment, as shown in FIG. **2** and FIG. **12**, the power supply system **100** also includes an AC input and protection circuit module **40** which includes an AC input sub-module **41**, an isolation switch sub-module **42**, a surge protection sub-module **43**, an electromagnetic interference sub-module **44**, and an input current limiting sub-module **45**. The AC input sub-module **41** is connected with the external power supply to receive the AC input of the external power supply; the isolation switch sub-module **42** is connected with the AC input sub-module **41** to protect the power supply system **100** to operate normally; the surge protection sub-module **43** is connected with the isolation switch sub-module **42** to prevent the damage caused by the surge voltage or current of the external power supply; the electromagnetic interference sub-module **44** is connected with the surge protection sub-module **43** to suppress electromagnetic interference; and the input current limiting sub-module **45** is connected with the electromagnetic interference sub-module **44** to limit the AC current of the external power supply.

[0118] In this embodiment, the AC input and protection circuit module **40** plays the main role of adapting to global voltage levels by setting a single three-phase wide voltage input connector in the AC input and protection circuit module **40**. In addition, the AC input and protection circuit module **40** is also set with a three-phase input voltage and current measurement and monitoring device to prevent many failures such as over-voltage, under-voltage, over-current and over-heat. The isolation switch sub-module **42** can quickly and safely disconnect the whole power supply system **100** in a failure. Besides, this module is also set with a surge protection sub-module **43**, an electromagnetic interference sub-module **44** and an input current limiting sub-module **45** to prevent surge, surge current, and electromagnetic interference.

[0119] In some embodiments, as shown in FIG. **2** and FIG. **12**, in this embodiment, the power supply system **100** also includes a DC/DC HF isolation transformer module **50** which is connected with the DC connecting bus module **20** and the voltage output circuit module **30** to regulate the DC voltage of the DC connecting bus module **20** to a DC voltage level required by the voltage output circuit module **30**.

[0120] In this embodiment, the DC/DC HF isolation transformer module mainly plays two roles. The first role is to isolate the primary and secondary edges with the HF transformer. The second role is to further regulate the DC bus voltage to a required DC voltage level. Wherein, in this embodiment, as shown in FIG. **2** and FIG. **12**, the DC/DC HF isolation transformer module can be set with DC/DC HF isolation transformers with corresponding quantity for the three-phase AC voltage output module **31**, the single-phase AC voltage output module **32** and the DC voltage output module **33**. In other embodiments, it is also possible to set a DC/DC HF isolation transformer with primary edge and many secondary edges, with no more restrictions here.

[0121] In addition, in this embodiment, a DC/DC HF isolation transformer is set to reduce the dimension and weight of the isolation transformer. The required size of the iron core and winding of the isolation transformer greatly reduces. The size of the whole isolation transformer is inversely

proportional to the frequency f . The relationship can be approximated as $A_c \times A_w \propto S / (B_m \times J \times f)$. Wherein, S refers to the output apparent power, B_m refers to the peak magnetic flux density, J refers to the current density of a conductor, A_c refers to the core area, and A_w refers to the window area. Therefore, the higher the frequency, the smaller the transformer. In this embodiment, the size of the isolation transformer is inversely proportional to its working frequency. If a 50 Hz power frequency transformer operates at 50 KHz, its volume can be reduced by 1000 times, that is, by 10 times in all dimensions, which can greatly reduce the volume and weight of the power supply system **100**.

[0122] Wherein, in this embodiment, refer to FIG. **13** to FIG. **15**. FIG. **13** illustrates a circuit structure schematic diagram of the first embodiment of the DC/DC HF isolation transformer in this application. FIG. **14** illustrates a circuit structure schematic diagram of the second embodiment of the DC/DC HF isolation transformer in this application. FIG. **15** illustrates a circuit structure schematic diagram of the third embodiment of the DC/DC HF isolation transformer in this application. In this embodiment, the DC/DC HF isolation transformer can be set as a circuit topological structure of any structure in FIGS. **13-15**. In other embodiments, the DC/DC HF isolation transformer also can have other topological forms, such as the dual active bridge DC/DC converter or the bidirectional current multiplication isolation converter. The DC/DC HF isolation transformer can increase the voltage grade and power output by cascade or parallel connection, with no more restrictions here.

[0123] In addition, refer to FIG. **16**. FIG. **16** illustrates a structure schematic diagram of the fourth embodiment of the DC/DC HF isolation transformer in this application. As shown in FIG. **16**, the DC/DC HF isolation transformer in this embodiment can be set as four topological structures. Wherein, Type A is a single-level structure, Type B and Type C are a two-level structure and Type D is a three-level structure. If the power supply system **100** is applied in medical devices, the DC/DC HF isolation transformer is preferably set as Type D. However, at a certain extent, Type A, Type B and Type C also can have the similar functions and realize the similar purposes, with no more restrictions here.

[0124] In some embodiments, as shown in FIG. **2** and FIG. **12**, in this embodiment, the power supply system **100** also includes an output monitoring module **60** and a grounding failure protection module **70**. The output monitoring module **60** is connected with the DC/DC HF isolation transformer module **50** to monitor the voltage and current outputted by the DC/DC HF isolation transformer module **50** and the grounding failure protection module **70** is connected with the voltage output circuit module **30** to monitor the grounding failure of the voltage output circuit module **30**.

[0125] In this embodiment, the output monitoring module **60** can monitor the output voltage and current in real time; and the grounding failure protection module **70** is set with a protection switch circuit breaker. In addition, in other embodiments, the grounding failure protection module **70** also can be set with grounding failure monitor, leakage current protector, insulation detector, short circuit protector and overload protector, with no more restrictions here.

[0126] The traditional power supply system **100** is protected by logic hardware circuits composed of switch components such as circuit breakers, relays, contactors, and suction coils. These hardware circuits have protection of over-voltage, over-current, under-voltage, over-load, over-heat, current leakage, phase loss, short circuit and grounding failures. The detection and protection circuit are mechanical. Therefore, they are complicated and are difficult to be realized. In addition, the volume and weight are high and the costs are high. Compared with the traditional power supply system **100**, this application takes the accurate measurement of voltage, current and temperature as the basis, the traditional power supply system **100** realizes the protection functions by signal acquisition, central microprocessor logic processing and electronic control switches. It can be seen that the protection logic is realized by software, the logic circuit is greatly simplified, the execution mechanism is also more energy-efficient and compact, and is characterized by small volume, light

weight and low costs.

[0127] In some embodiments, as shown in FIG. 2 and FIG. 12, in this embodiment, the power supply system **100** also includes a central protection and failure monitoring processing module **80** which includes a central management sub-module **81** and an external communication interface **82**. The central management sub-module **81** is used to monitor the status and safety of the power supply system **100** in real time and quickly and accurately execute protection tasks in a failure; the external communication interface **82** is connected with the central management sub-module **81** to realize the communication between the central management sub-module **81** and the external device.

[0128] Wherein, in this embodiment, the central management sub-module **81** is independent from controllers for other power converters in the system, but is established with communication with these controllers to complete tasks. In addition to the timing control for startup and shutdown, the central management sub-module **81** mainly monitors the status and safety of the whole power supply system **100** in real time and quickly and accurately executes protection tasks in a failure, such as input and output failures: over-voltage, over-current, under-voltage, under-current, overload, over-heat, phase loss, grounding failure and short circuit. At this time, the external communication interface **82** can ensure that the central management sub-module **81** can quickly acquire the external instructions so as to improve the management efficiency of the power supply system **100**.

[0129] In some embodiments, as shown in FIG. 2 and FIG. 12, in this embodiment, the power supply system **100** also includes an internal auxiliary power module **90** which is connected with the central management sub-module **81** to provide a safe internal power guarantee to the central management sub-module **81**.

[0130] In this embodiment, the internal auxiliary power module **90** mainly provides power supply to the central management sub-module **81**, protects the normal work, and provides a safe internal power guarantee. Particularly, it can work for a certain period in system power failure. Its safety requirement is higher than that of other power supplies. In addition, as shown in FIG. 2 and FIG. 12, the internal auxiliary power module **90** can be connected with the DC connecting bus **21**. When the power supply system **100** works normally, the internal auxiliary power module **90** can be charged by DC voltage from the DC connecting bus **21**.

[0131] Different from the related art, the power supply system **100** in the application includes a rectifying circuit module **10**, a DC connecting bus module **20**, and a voltage output circuit module **30**. The rectifying circuit module **10** is connected with an external power supply to correct and rectify the AC current of the external power supply to form a DC current; the DC connecting bus module **20** is connected with the rectifying circuit module **10** to store the DC current and output a DC voltage; and the voltage output circuit module **30** is connected with the DC connecting bus module **20** to convert the DC voltage of the DC connecting bus module **20** to at least any of a three-phase AC voltage, a single-phase AC voltage and a target DC voltage and output it. The power supply system **100** in the application takes the DC connecting bus module **20** as a transfer module to ensure stability and regulation of AC voltage and/or DC voltage outputted by the voltage output circuit module **30** without the influence of fluctuation on the AC side of the rectifying circuit module **10**. In addition, in the application, the voltage output circuit module **30** converts the DC voltage of the DC connecting bus module **20** to AC voltage and/or DC voltage, can replace the multiple-tap structure of traditional power supply systems **100** and can save costs.

[0132] In addition, the power supply system **100** in this application is controlled by many levels. Every level is isolated and is set with failure protection and interference noise filtering processing function. The failures and interference on the input and output sides will not spread. For example, if there are harmonic components in the input voltage, the control algorithm will have corresponding current harmonic suppression functions. In addition, the DC connecting bus **21** will have a low pass filter network to remove the voltage harmonic components to provide a stable DC

voltage to the latter level so as to isolate the interference. If there is a failure on the output side, the circuit will be in short circuit, and the power supply in the later stage will actively limit the current or the circuit will be cut to avoid influence of the early state. If the output is nonlinear AC load and the output current is not sinusoidal, the three-phase power factor correction circuit **11** in the early stage can correct the input current waveform to a sine wave so as to ensure high power factor and improve power supply quality. Besides, the power supply system **100** in this application meets the requirement of leakage current less than 5 mA and Class B conducted radiation. The application scope is wide. It can be used in medical devices.

[0133] In some embodiments, this application further proposes an electronic device. Refer to FIG. **17**. FIG. **17** illustrates a structure schematic diagram of an embodiment of the electronic device in this application. As shown in FIG. **17**, the electronic device **200** in this embodiment includes any power supply system **100** mentioned above.

[0134] Wherein, the electronic device **200** can be a medical device, and can be other electronic devices that require different voltage types and levels, with no more restrictions here.

[0135] The above content is only the embodiments in this application and constitutes no limitation to the scope of the patent in this application. Any equivalent structure or equivalent process transformation made by reference of the specification and accompanying drawings in this application, or direct or indirect application in other related technical fields are included in the protection scope of the patent of this application.

Claims

1. A power supply system, comprising: a rectifying circuit module, wherein the rectifying circuit module is connected with an external power supply to correct and rectify an AC current of the external power supply to form a DC current; a DC connecting bus module, wherein the DC connecting bus module is connected with the rectifying circuit module to store the DC current and output a DC voltage; and a voltage output circuit module, wherein the voltage output circuit module is connected with the DC connecting bus module to convert the DC voltage of the DC connecting bus module to at least any of a three-phase AC voltage, a single-phase AC voltage, or a target DC voltage and to output it.
2. A power supply system according to claim 1, wherein the DC connecting bus module comprises: a DC connecting bus, wherein the DC connecting bus is connected with the rectifying circuit module and the voltage output circuit module to store the DC current and output a DC voltage; and an input interface of the external power supply, wherein the input interface of the external power supply is connected with the DC connecting bus to connect with another external DC power supply.
3. A power supply system according to claim 1, wherein the rectifying circuit module comprises: a three-phase power factor correction circuit to correct and rectify the AC current of the external power supply to form a DC current; and a control circuit, wherein the control circuit is connected with the three-phase power factor correction circuit to realize the control of the three-phase power factor correction circuit by a digital signal processing method.
4. A power supply system according to claim 1, wherein the voltage output circuit module comprises: a three-phase AC voltage output module, wherein the three-phase AC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a three-phase AC voltage and output it; a single-phase AC voltage output module, wherein the single-phase AC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a single-phase AC voltage and output it; and a DC voltage output module, wherein the DC voltage output module is connected with the DC connecting bus module to convert the DC voltage to a target DC voltage and output it.
5. A power supply system according to claim 4, wherein the DC connecting bus module comprises a first DC connecting bus, a second DC connecting bus and an input interface of the external power

supply, and the rectifying circuit module comprises a first three-phase power factor correction circuit, a second three-phase power factor correction circuit, and a control circuit; and wherein the first three-phase power factor correction circuit is connected with the first DC connecting bus, and the three-phase AC voltage output module and the single-phase AC voltage output module are connected with the first DC connecting bus; and the second three-phase power factor correction circuit is connected with the second DC connecting bus and the DC voltage output module is connected with the second DC connecting bus.

6. A power supply system according to claim 4, wherein the three-phase AC voltage output module comprises: a three-phase DC to AC converter, wherein the three-phase DC to AC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a three-phase AC voltage; a first AC filter, wherein the first AC filter is connected with the three-phase DC to AC converter to filter the three-phase AC voltage; and a three-phase AC output circuit, wherein the three-phase AC output circuit is connected with the first AC filter to output the three-phase AC voltage.

7. A power supply system according to claim 4, wherein the single-phase AC voltage output module comprises: a single-phase DC to AC converter, wherein the single-phase DC to AC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a single-phase AC voltage; a second AC filter, wherein the second AC filter is connected with the single-phase DC to AC converter to filter the single-phase AC voltage; and a single-phase AC output circuit, wherein the single-phase AC output circuit is connected with the second AC filter to output a single-phase AC voltage.

8. A power supply system according to claim 4, wherein the DC voltage output module comprises: a DC to DC converter, wherein the DC to DC converter is connected with the DC connecting bus to convert the DC voltage of the DC connecting bus to a target DC voltage; a DC filter, wherein the DC filter is connected with the DC to DC converter to filter the target DC voltage; and a DC output circuit, wherein the DC output circuit is connected with the DC filter to output the target DC voltage.

9. A power supply system according to claim 1, wherein the power supply system also comprises an AC input and protection circuit module comprising: an AC input sub-module, wherein the AC input sub-module is connected with the external power supply to receive the AC input of the external power supply; an isolation switch sub-module, wherein the isolation switch sub-module is connected with the AC input sub-module to protect the power supply system to operate normally; a surge protection sub-module, wherein the surge protection sub-module is connected with the isolation switch sub-module to prevent damage caused by the surge voltage or current of the external power supply; an electromagnetic interference sub-module, wherein the electromagnetic interference sub-module is connected with the surge protection sub-module to suppress electromagnetic interference; and an input current limiting sub-module, wherein the input current limiting sub-module is connected with the electromagnetic interference sub-module to limit the AC current of the external power supply.

10. A power supply system according to claim 1, wherein the power supply system also comprises a DC/DC HF isolation transformer module, and wherein the DC/DC HF isolation transformer module is connected with the DC connecting bus module and the voltage output circuit module to regulate the DC voltage of the DC connecting bus module to a DC voltage level required by the voltage output circuit module.

11. A power supply system according to claim 10, wherein the power supply system also comprises an output monitoring module and a grounding failure protection module; and the output monitoring module is connected with the DC/DC HF isolation transformer module to monitor the voltage and current outputted by the DC/DC HF isolation transformer module and the grounding failure protection module is connected with the voltage output circuit module to monitor the grounding failure of the DC voltage outputted by the voltage output circuit module.

12. A power supply system according to claim 1, wherein the power supply system also comprises a central protection and failure monitoring processing module comprising: a central management sub-module to monitor status and safety of the power supply system in real time and quickly and accurately execute protection tasks in a failure; and an external communication interface, wherein the external communication interface is connected with the central management sub-module to realize communication between the central management sub-module and an external device.

13. A power supply system according to claim 12, wherein the power supply system also comprises an internal auxiliary power module, and the internal auxiliary power module is connected with the central management sub-module to provide a safe internal power guarantee to the central management sub-module.

14. An electronic device, wherein the electronic device comprises a power supply system, comprising: a rectifying circuit module, wherein the rectifying circuit module is connected with an external power supply to correct and rectify an AC current of the external power supply to form a DC current; a DC connecting bus module, wherein the DC connecting bus module is connected with the rectifying circuit module to store the DC current and output a DC voltage; and a voltage output circuit module, wherein the voltage output circuit module is connected with the DC connecting bus module to convert the DC voltage of the DC connecting bus module to at least any of a three-phase AC voltage, a single-phase AC voltage, or a target DC voltage and to output it.
